



Validation of CanopySat Bioclimatic Envelope Assessment (BEA)

Comparison Against Published MaxEnt Studies — 5 Global Forest Biomes — SSP2-4.5 (2050)

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ABSTRACT

This report validates the CanopySat Bioclimatic Envelope Assessment (BEA) methodology for predicting forest species climate suitability under 2050 climate projections (SSP2-4.5). BEA results across five global biomes are compared against published MaxEnt species distribution models. Species suitability trends predicted by CanopySat BEA are consistent with MaxEnt findings for general directional trends across tropical, temperate and boreal biomes. Climate projections use a 5-model CMIP6 ensemble (ACCESS-CM2, MIROC6, MPI-ESM1-2-HR, GFDL-ESM4, BCC-CSM2-MR) under SSP2-4.5. Species occurrence data sourced from GBIF (CC BY 4.0).

METHODOLOGY

The BEA method estimates species thermal tolerance from global GBIF occurrence data (100 records/species) using the formula $T = 27 - |\text{lat}| \times 0.52 \text{ degC}$. Suitability scores are calculated by comparing projected 2050 temperature (5-model CMIP6 ensemble mean, SSP2-4.5) against each species thermal tolerance range. Penalties are applied proportionally when 2050 temperature exceeds the tolerance range. Classification thresholds: Suitable (score ≥ 0.70), Moderate risk (0.40-0.69), At risk (< 0.40). Current climate from WorldClim V1 (BIO01, BIO12).

RESULTS BY BIOME

Amazon Rainforest — Tropical

Current temperature: 26.6°C | 2050 projection: 29.2°C (+2.5°C) | Precipitation: 2612 mm → 2108 mm/year (-19.3%)

Metric	Value
Species analyzed	15
Suitable (score ≥ 0.70)	9
Moderate risk (0.40-0.69)	6
At risk (< 0.40)	0

MaxEnt reference: Mixed — some species decline, others stable under +2.5°C

Source: Murcia-Garcia et al. 2025, Forests; Vicari 2025, UFV Brazil

BEA vs MaxEnt: **CONSISTENT with MaxEnt**

Congo Basin — Tropical

Current temperature: 24.7°C | 2050 projection: 27.3°C (+2.6°C) | Precipitation: 1885 mm → 1751 mm/year (-7.1%)

Metric	Value
Species analyzed	15
Suitable (score ≥ 0.70)	14
Moderate risk (0.40-0.69)	1
At risk (< 0.40)	0

MaxEnt reference: Mostly stable under moderate warming (+2.6°C)

Source: Pan et al. 2011, Science; FAO FRA 2020

BEA vs MaxEnt: CONSISTENT with MaxEnt

Black Forest — Temperate

Current temperature: 8.4°C | 2050 projection: 9.1°C (+0.7°C) | Precipitation: 1039 mm → 1024 mm/year (-1.4%)

Metric	Value
Species analyzed	15
Suitable (score >= 0.70)	10
Moderate risk (0.40-0.69)	4
At risk (< 0.40)	1

MaxEnt reference: Temperate species show moderate stress; Fagus sylvatica at risk above +2°C

Source: Hanewinkel et al. 2013, Nature Climate Change

BEA vs MaxEnt: CONSISTENT with MaxEnt

Borneo — Tropical SE Asia

Current temperature: 21.9°C | 2050 projection: 24.8°C (+2.9°C) | Precipitation: 3639 mm → 4169 mm/year (14.6%)

Metric	Value
Species analyzed	11
Suitable (score >= 0.70)	11
Moderate risk (0.40-0.69)	0
At risk (< 0.40)	0

MaxEnt reference: Dipterocarp species mostly suitable under +2.9°C SSP245

Source: Slik et al. 2010, Global Ecology & Biogeography

BEA vs MaxEnt: CONSISTENT with MaxEnt

Boreal Canada — Boreal

Current temperature: -1.7°C | 2050 projection: 1.5°C (+3.2°C) | Precipitation: 485 mm → 456 mm/year (-6.1%)

Metric	Value
Species analyzed	15
Suitable (score >= 0.70)	14
Moderate risk (0.40-0.69)	1
At risk (< 0.40)	0

MaxEnt reference: Boreal species show expansion northward; mostly stable under +3.2°C

Source: Pan et al. 2011, Science; IPCC AR6 Chapter 2

BEA vs MaxEnt: CONSISTENT with MaxEnt

SUMMARY — BEA vs MaxEnt CONSISTENCY

Forest Zone	Temp +°C	Suitable	Moderate	At risk	MaxEnt Consistency
Amazon Rainforest	+2.5°C	9	6	0	Consistent
Congo Basin	+2.6°C	14	1	0	Consistent
Black Forest	+0.7°C	10	4	1	Consistent
Borneo	+2.9°C	11	0	0	Consistent
Boreal Canada	+3.2°C	14	1	0	Consistent
ALL ZONES					5/5 Consistent

LIMITATIONS

The BEA method provides directional trends consistent with MaxEnt but is not equivalent to full MaxEnt modeling. Limitations include: temperature estimated from latitude (not measured), no altitude/soil/competition factors, 100 GBIF occurrences per species (may be insufficient for rare species), and single SSP2-4.5 scenario. For species-specific conservation decisions, full MaxEnt modeling with field validation is recommended.

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